

Discrete Entropic Gravity on a Cubic Graph: Emergent Newtonian and MOND Regimes with Λ -Locked Screening

Morten Magnusson 

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Abstract

We construct a discrete gravity operator defined on a cubic graph with entropic flux weighting between nodes. The operator reproduces (i) Newtonian $1/r^2$ scaling in the strong-field (high-acceleration) regime, (ii) MOND-like $1/r$ scaling in the low-acceleration regime, (iii) a transition scale set by a bulk entropic reservoir proportional to Λ , (iv) broken superposition, and (v) an external field effect (EFE).

No continuum metric or General Relativity equations are assumed. The theory is defined purely on graph topology. Systematic convergence and kill-tests are performed. A non-trivial prefactor $C \approx 2.32$ emerges independent of spherical vs cubic binning, indicating a discrete renormalization of the infrared response. This implies an effective acceleration scale $a_{0,\text{eff}} = C^2 a_0 \approx 5.4 a_0$, which deviates from the standard MOND value and must be accounted for in any phenomenological comparison.

1 Introduction

Modified gravity models often introduce phenomenological interpolating functions to recover MOND-like behaviour in weak-field regimes [1, 2]. In contrast, we construct a discrete operator on a cubic graph and show that Newtonian and MOND-like regimes emerge from the same entropic flux functional.

The transition scale is set by a bulk entropic term proportional to Λ , producing a screening length

$$L_\Lambda \propto \Lambda^{-1/2} \quad (1)$$

and an associated acceleration scale

$$a_0 \propto c^2 \sqrt{\Lambda}. \quad (2)$$

This paper is a technical note establishing the operator and its convergence properties. It does not claim equivalence to MOND or to any cosmological model; it reports what the graph-based operator produces under controlled numerical conditions.

2 Graph Construction

We define a cubic lattice with N^3 nodes. Each interior node i connects to its six nearest neighbours; boundary nodes connect to fewer.

Let Φ_i denote the potential at node i . The discrete gradient magnitude between nodes is

$$|\nabla \Phi|_{ij} = \frac{|\Phi_i - \Phi_j|}{a}, \quad (3)$$

where a is lattice spacing.

3 Discrete AQUAL Operator

Following the AQUAL formulation [3], we define weights

$$w_{ij} = \mu \left(\frac{|\Phi_i - \Phi_j|}{a_0 a} \right), \quad (4)$$

with interpolating function

$$\mu(x) = \frac{x}{\sqrt{1+x^2}}. \quad (5)$$

The discrete field equation is

$$\sum_{j \sim i} w_{ij} (\Phi_j - \Phi_i) = 4\pi G \rho_i. \quad (6)$$

This is the graph analogue of AQUAL.

4 Bulk Entropic Term and Λ -Locking

Motivated by entropic gravity approaches [4], we add a bulk entropic reservoir term that couples the local entropy S_i at each node to a volume-averaged reference entropy S_V :

$$F_{\text{bulk}} = \frac{1}{2} \mu_\Lambda (S_i - S_V)^2, \quad (7)$$

with

$$\mu_\Lambda \propto \Lambda. \quad (8)$$

This produces a Yukawa-like screening scale

$$L_\Lambda \sim \frac{1}{\sqrt{\Lambda}}, \quad (9)$$

and therefore

$$a_0 = \frac{c^2}{L_\Lambda} \propto c^2 \sqrt{\Lambda}. \quad (10)$$

The cosmological constant value is taken from [5].

5 Kill Tests

We perform five kill-tests (KTs). Each is defined such that failure would falsify the operator. The field equation is solved iteratively via damped nonlinear relaxation until $\|\Delta\Phi\|_\infty < 10^{-4}$. All numerical results are from the solver described in [8].

5.1 KT1: Newton and MOND Recovery

The operator must recover $g \propto r^{-2}$ in the Newtonian regime and $g \propto r^{-1}$ in the deep-MOND regime. We measure log-log slopes in the inner ($3 < r/a < 5$) and outer ($r > r_{\text{MOND}}$) regions for different values of a_0 (in lattice units):

Regime	Measured slope	Expected
Newton ($a_0 = 0.001$)	-2.00	-2.0
MOND ($a_0 = 2.0$, outer)	-0.99	-1.0
MOND ($a_0 = 0.5$, outer)	-1.01	-1.0

Result: **PASS**. Both asymptotic regimes are recovered to within numerical precision.

5.2 KT2: Prefactor Convergence

In the deep-MOND regime, the standard prediction is $g = \sqrt{a_0 g_N}$. On the graph, we measure

$$C = \frac{g_{\text{AQUAL}}}{\sqrt{a_0 g_N}}. \quad (11)$$

N	C_{cubic}	C_{sphere}
21	2.443	2.433
31	2.354	2.349
41	2.320	2.321

Convergence suggests $C \rightarrow 2.32$ as $N \rightarrow \infty$. This prefactor is independent of binning geometry (cubic vs spherical), indicating it is intrinsic to the discrete topology.

Interpretation. $C \neq 1$ means the graph operator does not reproduce standard MOND. The effective deep-MOND relation is

$$g_{\text{graph}} = C \sqrt{a_0 g_N}, \quad (12)$$

which is equivalent to an effective acceleration scale

$$a_{0,\text{eff}} = C^2 a_0 \approx 5.4 a_0. \quad (13)$$

This is a discrete renormalization effect. Any comparison with observed rotation curves or galaxy dynamics must use $a_{0,\text{eff}}$, not a_0 . We do not claim this matches phenomenological MOND; we report what the operator produces.

5.3 KT3: Mass Scaling

In standard MOND, the transition radius (defined as the radius where $g = a_0$) scales as $r_{\text{trans}} \propto M^{1/2}$. On the graph, we measure

$$r_{\text{trans}} \propto M^{0.18 \pm 0.03}. \quad (14)$$

This is substantially weaker than the MOND expectation (0.5). The weak mass dependence indicates that Λ -locking dominates: the transition is set primarily by the bulk entropic scale L_Λ , not by the local mass. This is a structural consequence of the Λ -proportional reservoir term and represents a clear departure from standard MOND phenomenology.

5.4 KT4: Broken Superposition

For two equal point masses separated by $d = 8a$, the nonlinear superposition violation is measured as

$$\delta\Phi/\Phi_{\text{max}} = 13.7\%, \quad (15)$$

where $\delta\Phi = \Phi_{\text{joint}} - (\Phi_1 + \Phi_2)$ is the difference between the joint solution and the sum of individual solutions.

The violation has smooth spatial structure (not noise), confirming it arises from the nonlinear μ -function. This is expected from any AQUAL-type operator and constitutes a consistency check.

5.5 KT5: External Field Effect

After radial projection, enhancement decreases monotonically with external field strength, indicating partial EFE consistent with the nonlinear operator. The EFE is qualitative in this test; a quantitative measurement requires galaxy-pair boundary conditions.

6 Cosmological Stability Check

Although this note concerns the discrete operator in the quasi-static regime, we report a minimal consistency check against cosmological observables to verify the operator does not produce catastrophic growth enhancement.

Using the background modification derived from the same Λ -locked coupling (with $\beta = 0.08$, background channel only, no Poisson modification), we solve the linear growth ODE

$$\ddot{\delta} + 2H(a)\dot{\delta} - \frac{3}{2}\Omega_m H_0^2(1+z)^3\delta = 0 \quad (16)$$

with the modified $H(a)$, and compare against BOSS DR12 $f\sigma_8$ measurements [6]:

	σ_8	$f\sigma_8(z=0.51)$	ε_0
ΛCDM	0.811	0.473	—
This work (background)	0.805	0.463	0.8%

Here ε_0 is the total growth suppression relative to ΛCDM . The model does not produce runaway structure growth: σ_8 is suppressed by 0.7%, consistent with the direction required by the S_8 tension. This is a stability check, not a fit. The perturbation-level $\mu(a) < 1$ channel remains disabled in this test; all growth modification enters through enhanced Hubble friction in the background channel only.

7 Discussion

The operator reproduces Newtonian and MOND-like asymptotics, Λ -locked transition, broken superposition, and external field sensitivity. Several features distinguish it from standard MOND:

- The prefactor $C \approx 2.32$ implies $a_{0,\text{eff}} \approx 5.4 a_0$. This is not a numerical artifact; it persists under resolution refinement and across binning geometries. It represents discrete renormalization of the infrared response.
- The mass-scaling exponent (0.18) is substantially below the MOND expectation (0.50), reflecting Λ -dominance over local mass in setting the transition radius.
- Both deviations are structural consequences of the graph topology and Λ -locked bulk term. Whether they improve or worsen agreement with galaxy-scale data remains to be tested.

For broader context on modified gravity approaches, see [7]. The full numerical implementation, convergence data, and kill-test suite are available in [8].

8 Conclusion

We have constructed a purely graph-based gravity operator that reproduces Newtonian and MOND-like behaviour without assuming a continuum metric. The transition scale emerges from Λ -proportional bulk entropic coupling.

The operator produces two quantitative departures from standard MOND: a convergent prefactor $C \approx 2.32$ and weak mass scaling ($r_{\text{trans}} \propto M^{0.18}$). Both are intrinsic to the discrete topology and must be confronted with observations before any phenomenological claim can be made.

Future work will examine: (i) calibration against observed rotation curves using $a_{0,\text{eff}}$, (ii) full perturbation-level cosmological integration, and (iii) the continuum limit of the prefactor.

References

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